MTH 335(A) Dr. James Ashe

Homework 4 Due date: Friday 19^{th} April, 2013 **Solutions**

Homework Problems. *Solutions should be written/typed in numerical order and clearly labeled. Unsolved problems should be included but left blank.*

1. Let $a, b, c \in \mathbb{Z}$. Prove the following:

a. If $a|b$ and $b|c$ then $a|c$.

Proof. $a|b \Rightarrow b = na$ for some $n \in \mathbb{Z}$, and $b|c \Rightarrow mb = m(na) = c$ for some $m \in \mathbb{Z}$. Thus $a|c$ □

b. If $a|b$ and $a|c$ then $a|b \pm c$.

Proof. Let $a|b$ and $a|c$. So then $b = an$ and $c = am$ for some $n, m \in \mathbb{Z}$. Thus $b \pm c = an \pm am = a(n \pm m)$ which implies that $a|b \pm c$. □

2. Show that the set of integers, $\mathbb{Z}$, is a group under addition.

Proof. Let $a, b, c \in \mathbb{Z}$. Then $a + b \in \mathbb{Z}$ (closure); $a + 0 = 0 + a = a$ so $a = e$ (identity); $a - a = 0$ so $a^{-1} = -a \in \mathbb{Z}$ (inverses), and $(a + b) + c = a + (b + c)$ by associativity of real numbers (associativity). So then $\mathbb{Z}$ is a group under addition. □

3. Prove that the set of even integers is a group under addition.

Proof. Let $a, b \in 2\mathbb{Z}$ then $a = 2n$ and $b = 2m$ for some $m, n \in \mathbb{Z}$. $a + b = 2n + 2m = 2(n + m) \in 2\mathbb{Z}$ so it is closed. The identity is $0 = 2 \cdot 0 \in 2\mathbb{Z}$, $a^{-1} = -a = -2n \in 2\mathbb{Z}$, and it is associative by the associativity of $\mathbb{Z}$. □

4. Show that the following are *not* groups under the given operation.

a. $\mathbb{Z}$ under multiplication.

Proof. $2 \in \mathbb{Z}$, but $2^{-1} \notin \mathbb{Z}$ since $2^{-1}2 = 1 \Rightarrow 2^{-1} = \dfrac{1}{2}$. □

b. $S = \left\{ z | z \in \mathbb{Z} \text{ and } z = \sqrt{5} \right\}$ under any operation.

Proof. $\sqrt{5} \notin \mathbb{Z}$ so then $S = \emptyset$. So S is not a group. □

c. $\mathbb{Q}$, the set of rationals under multiplication.

Proof. $0 \in \mathbb{Q}$, but 0 has no multiplicative inverse. □

d. The set of odd integers under addition.

Proof. 3 and 5 are odd, but $3 + 5 = 8 = 2 \cdot 4$ which is not odd. So it is not closed. □

5. Show that $\{1, 2, 3\}$ under multiplication modulo 4 is *not* a group but that $\{1, 2, 3, 4\}$ under multiplication modulo 5 is a group. (Example: $2 \cdot 3 = 6$ and $6 = 2 \bmod 4$ so $[6] = [2]$ and thus $2 \cdot 3$ *is* in $\{1, 2, 3\}$).

Proof. $2 \cdot 2 = 0 \bmod 4$ and $0 \notin \{1, 2, 3\}$ so it is not a group.

$\{1, 2, 3, 4\}$:

	1	2	3	4
1	1	2	3	4
2	2	4	1	3
3	3	1	4	2
4	4	1	2	3

So it is closed, the identity is 1, each element has an inverse, and associativity follows by the associativity of modular multiplication. □

6. Prove that the set of even integers is a group under addition. (I mistakenly repeated problem 3. For a more challenging problem you might try showing that integers of n multiples for $n \in \mathbb{Z}^+$ are a group under addition)

Proof. Let $n\mathbb{Z} = \{x \in \mathbb{Z} \text{ such that } n|x\}$ for some integer n. Let $a, b \in n\mathbb{Z}$. Then $a = nx$ and $b = ny$ for some $x, y \in \mathbb{Z}$. $a + b = nx + ny = n(x + y)$ so $n|a + b$, and $n\mathbb{Z}$ is closed. The identity is 0, $a^{-1} = -a = -nx$ which is divisible by n, and associativity follows from associativity of $\mathbb{Z}$. □

7. Let $S = \left\{ a + b\sqrt{5} | \, a, b \text{ are not both zero and } a, b \in \mathbb{Q} \right\}$. Prove that S is a group under multiplication. (Hint: rationalize the denominator to show that the inverse of $a + b\sqrt{5}$ is in S.)

Proof. Let $x, y \in S$ so then $x = a + b\sqrt{5}$ and $y = c + d\sqrt{5}$ for some $a, b, c, d \in \mathbb{Q}$. $xy = ac + (b + d)\sqrt{5} + 5bd = ab + 5bd + (a + b)\sqrt{5}$ which is in S as $\mathbb{Q}$ is closed under multiplication and addition. 1 is the identity. $x^{-1} = \dfrac{1}{a + b\sqrt{5}} \dfrac{(a - b\sqrt{5})}{(a - b\sqrt{5})} = \dfrac{a - b\sqrt{5}}{a^2 - 5b} = \dfrac{a}{a^2 - 5b} + \dfrac{-b}{a^2 - 5b}\sqrt{5} \in S$. Associativity follows by the associativity of $\mathbb{R}$. □

8. Prove that the set of all 2×2 matrices with determinant 1 and entries from $\mathbb{R}$ is a group under matrix multiplication.

Proof. Call this set SL. Let $A, B \in SL$. The product AB will be a 2×2 matrix anddet$(A)\det(B) = \det(AB)$. Thus $AB \in SL$. The identity matrix has determinant 1 so it is in SL. Since $\det(A) \neq 0$ there exists an inverse matrix A^{-1} and $\det(A^{-1}) = \dfrac{1}{\det(A)} = 1$. Thus $A^{-1} \in SL$. Matrix multiplication is associative. So SL is a group.

This group is called the special linear group of degree 2, denoted $\mathrm{SL}(2, \mathbb{R})$. In general $n \times n$ matrices with determinants of 1 are groups called the special linear group of degree n, denoted $\mathrm{SL}(n, \mathbb{R})$. □

9. Suppose that if G is a group such that if $a, b, c \in G$ and $ab = ca$ then $b = c$. Prove that G is an abelian group, i.e., that $ab = ba$ for all $a, b \in G$. Hint: consider the element $aba = aba$.

Proof. Let $a, b, c \in G$ and suppose that $ab = ca$ implies $b = c$. G is a group so then $a(ba) = (ab)a$ by associativity. By the assumption $a(ba) = (ab)a$ implies that $ab = ba$ so then G is abelian. □

10. Prove that G is an abelian group if and only is $(ab)^{-1} = a^{-1}b^{-1}$ for all $a, b \in G$.

Proof. ($\Rightarrow$) Suppose that G is abelian. Then $(ab)^{-1} = (ba)^{-1}$. $(ba)^{-1}ba = e \Rightarrow (ba)^{-1} = a^{-1}b^{-1}$ as required.
($\Leftarrow$) Suppose that $(ab)^{-1} = a^{-1}b^{-1}$. Similarly to above $(ab)^{-1} = b^{-1}a^{-1}$ So then $ab(ab)^{-1} = e \Rightarrow ab(a^{-1}b^{-1}) = e \Rightarrow abab^{-1}ba = ba \Rightarrow ab = ba$ □

Theorem. The Fundamental Theorem of Arithmetic. *Every integer greater than 1 is either a prime or the unique product of primes.*

Definition. Let a and n be integers with $n > 0$. The *congruence class of a modulo n* (denoted $[a]$) is the set of integers that are congruent to a modulo n, i.e.,

$$[a] = \{b | b \in \mathbb{Z} \text{ and } b = a \bmod n\}$$

For example, the congruence class of 3 modulo 5 is $[3] = \{\ldots, -2, 3, 8, 13, \ldots\}$.

Definition. The set of all congruence classes modulo n is denoted $\mathbb{Z}_n$ (read "Z mod n").

- $|\mathbb{Z}_n| = n$.
- Addition in $\mathbb{Z}_n$ is defined as $[a] + [b] = [a + b]$.
- Multiplication in $\mathbb{Z}_n$ is defined as $[a] \cdot [b] = [a \cdot b]$.

For example, $\mathbb{Z}_4 = \{[0], [1], [2], [3]\}$ for convenience we write $\mathbb{Z}_4 = \{0, 1, 2, 3\}$. $2 + 3 = 5$ and $5 = 1 \bmod 4$ so $[5] = [1]$.

Definition. A subset H of a group G is a *subgroup* of G is H is also a group under the operation in G.